University of Debrecen
Faculty of Engineering
Institute of Vehicle Engineering and Mechatronics
Department of Electrical Engineering and Mechatronics

Modelling DC Motor(1st Assignment)

ELECTRICAL MACHINES AND DRIVES

2025/2026/2. semester

Carried out by:

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Program:	Mechatronic Engineering BSc
Date:	2026. 04. 11.

Mandatory Tasks

Repeat the analysis of the catalogue data described above for the motor of your choice (everyone should choose a brushed pm DC motor, e.g. from the Faulhaber or Maxon website. Similar calculations performed on the same motor will not be accepted.)

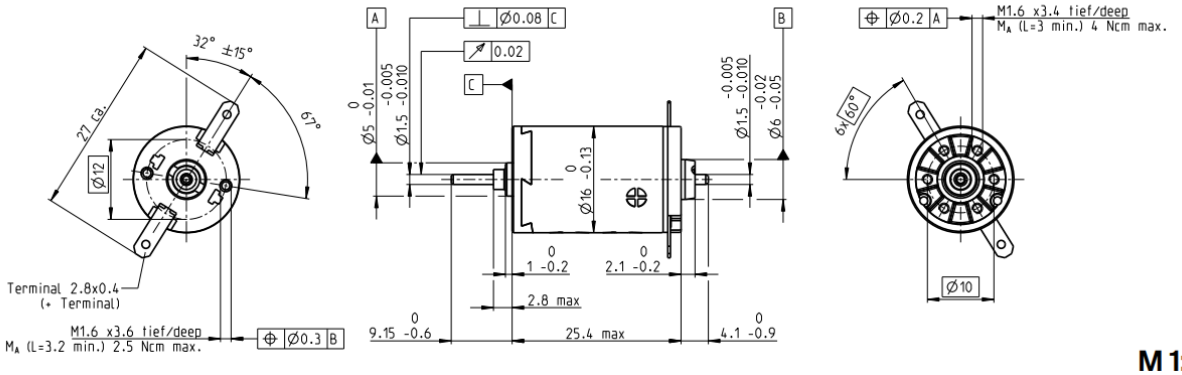
Define two transfer functions relevant to the motor model, using terminal voltage as input and angular velocity/current as output. Describe in detail the rearrangement of equations and provide a thorough derivation of the transfer functions. Additionally, validate the motor model's transfer function using MATLAB

Using Matlab/Simulink, create models of the DC motor, substitute the parameters from the catalogue data, and compare the results (unit jump to input signal) with those calculated in the previous section.
Make 3 DC motor models in Matlab/Simulink:

- Model using DC Machine block
- Block diagram model
- Model using Transfer Function block (in this case you cannot add an external load to the motor)

Compare the datasheet parameters with the results obtained from the simulation. (Nominal voltage ⟶ no-load, nominal load, stall torque ⟶ angular velocity? & current?)

Summarize the tasks that have been completed.



M 1:1

Catalogue Data for DC-max 16 S (24 V Model)

Parameter Description	Symbol	Value
Values at nominal voltage		
Nominal voltage	U_{an}	24 V
Nominal power	P_{mn}	4.7 W
No load speed	Ω_{nl}	9920 rpm
No load current	I_{anl}	16.8 mA

Nominal speed	Ω_n	6580 rpm
Nominal torque (max. continuous torque)	M_n	4.76 mNm
Nominal current (max. continuous current)	I_n	0.227 A
Stall torque	M_{rz}	14.4 mNm
Stall current	I_{rz}	0.64 A
Max. efficiency	η	71%
Characteristics		
Terminal resistance	R_a	37.5 Ω
Terminal inductance	L_a	1.37 mH
Torque constant	K_m	22.5 mNm/A
Speed constant	K_v	424 rpm/V
Speed/torque gradient		707 rpm/mNm
Mechanical time constant	T_m	8.57 ms
Electrical time constant (calculated)	T_E	0.03653 ms
Rotor inertia	J_m	1.16 gcm^2
Thermal data		
Thermal resistance housing-ambient	$R_{th:h,a}$	29.8 K/W
Thermal resistance winding-housing	$R_{th:w,h}$	5.5 K/W
Thermal time constant winding	$T_{th:w}$	5.35 s
Thermal time constant motor	$T_{th:m}$	288 s
Ambient temperature		-30°C to +85 °C
Max. winding temperature		+125 °C
Mechanical data (ball bearings)		
Max. speed		11000 rpm
Axial play		0.05-0.15 mm
Radial play		0.025 mm
Max. axial load (dynamic)		2.2 N
Max. force for press fits (static)		30 N
(static, shaft supported)		200 N
Max. radial load		7.8 N (5 mm from flange)
Other specifications		
Number of pole pairs		1
Number of commutator segments		7

Weight of motor		23.1 g

Solution

Definition of Motor Parameters

The first step is to define the data: This is a permanent magnet brushed graphite DC motor. The values of R_a, L_a and J_m (the subscript m expresses that J_m it is for the motor only; the load inertia is added to it) can be read directly, and the units of measurement can be made uniform.

$$R_a = 37.5\ \Omega, \quad L_a = 1.37$$

The torque constant (K_m) and speed constant (K_v) are linked by the magnetic flux constant ($k\phi$).

$$K_m = 22.5\ mNm/A = 0.0225$$

$$k\phi = K_m = 0.0225\ Vs/rad$$

Checking via speed constant:

$$K_v = 424\ \frac{rpm}{V} = 44.40$$

Viscous Friction (B_m)

B_m is calculated from the no-load current (I_{anl}):

$$B_m = \frac{(k\phi)^2 I_{anl}}{(U_{an} - R_a I_{anl})} = \frac{0.0225^2 \cdot 0.0168}{24 - 37.5 \cdot 0.0168} = 3.639 \cdot 10^{-7}\ Nms$$

Validation and Characteristics

The speed/torque gradient is:

$$\frac{R_a}{(k\phi)^2} = \frac{37.5}{0.0225^2} = 74074.07\ \frac{rad}{sNm} = 707.36$$

Calculated no-load speed:

$$\Omega_{nl} = K_v(U_{an} - R_a I_{anl}) = 424 \cdot (24 - 37.5 \cdot 0.0168) = 9908.8\ rpm$$

Calculated stall current and torque:

$$I_{rz} = \frac{U_{an}}{R_a} = \frac{24}{37.5} = 0.64\ A, \quad M_{rz} = K_m I_{rz} = 0.0225 \cdot 0.64 = 14.4$$

Time Constants (Defining $T_e, T_m, T_1,$ and T_2)

We distinguish between the theoretical time constants (T_e, T_m) and the actual time constants derived from the system poles (T_1, T_2).

Theoretical Time Constants:

- **Electrical:**
  $$T_e = \frac{L_a}{R_a} = \frac{0.00137}{37.5} = 0.0365\ ms$$
- **Mechanical:**
  $$T_m = \frac{J_m R_a}{(k\phi)^2} = \frac{1.16 \cdot 10^{-7} \cdot 37.5}{0.0225^2} = 8.59\ ms$$

Actual Time Constants from the Dynamic Model (T_1, T_2):

When solving the characteristic equation

$J_m L_a s^2 + (B_m L_a + J_m R_a)s + (k\phi)^2 + B_m R_a = 0$, We find two roots (poles).

- The high-frequency pole ($s_2 \approx -27254$) gives the electromagnetic time constant T_1:
  $$T_1 = \frac{1}{|s_2|} = 0.0367\ ms$$
- The low-frequency pole ($s_1 \approx -120.18$) gives the electromechanical time constant T_2:
  $$T_2 = \frac{1}{|s_1|} = 8.32\ ms$$

Comparison:

$$T_e = 0.0365\ ms \approx T_1 = 0.0367$$

$$T_m = 8.59\ ms \approx T_2 = 8.32$$

Note that T_2 (the actual mechanical response time) is slightly smaller than T_m because viscous friction (B_m) helps the motor reach steady state faster.

Transfer Functions

To derive the transfer functions, we start with the basic differential equations for a DC motor in

the time domain:

1. **Electrical (Kirchhoff's Voltage Law):**

   $$U_{an}(t) = R_a \cdot i(t) + L_a \frac{di(t)}{dt} + k\phi \cdot \Omega(t)$$

2. **Mechanical (Newton's Second Law):**

   $$M_m(t) = J_m \frac{d\Omega(t)}{dt} + B_m \cdot \Omega(t) + M_{load}(t)$$

   (Note: For the transfer function derivation, we assume no external load, so $M_{load}(t) = 0$*. The motor torque is* $M_m(t) = k\phi \cdot i(t)$*.*

Step 1: Convert to Laplace Domain (s-domain)

Assuming zero initial conditions, we take the Laplace transform of both equations:

1. $U_{an}(s) = R_a \cdot I(s) + sL_a \cdot I(s) + k\phi \cdot \Omega(s)$
2. $k\phi \cdot I(s) = sJ_m \cdot \Omega(s) + B_m \cdot \Omega(s)$

Step 2: Deriving the Angular Velocity Transfer Function

$$\frac{\Omega(s)}{U_{an}(s)}$$

First, rearrange the electrical equation to solve for $I(s)$:

$$I(s) = \frac{U_{an}(s) - k\phi \cdot \Omega(s)}{R_a + sL_a}$$

Next, substitute this $I(s)$ into the mechanical equation:

$$k\phi \left(\frac{U_{an}(s) - k\phi \cdot \Omega(s)}{R_a + sL_a} \right) = \Omega(s)(sJ_m + B_m)$$

Multiply both sides by $(R_a + sL_a)$ to clear the denominator:

$$k\phi \cdot U_{an}(s) - (k\phi)^2 \cdot \Omega(s) = \Omega(s)(sJ_m + B_m)(R_a + sL_a)$$

Group all $\Omega(s)$ terms on one side:

$$k\phi \cdot U_{an}(s) = \Omega(s)[(sJ_m + B_m)(sL_a + R_a) + (k\phi)^2]$$

Expand the brackets and divide to get the final transfer function:

$$\frac{\Omega(s)}{U_{an}(s)} = \frac{k\phi}{J_m L_a s^2 + (J_m R_a + B_m L_a)s + B_m R_a + (k\phi)^2}$$

Step 3: Deriving the Armature Current Transfer Function

$$\frac{I(s)}{U_{an}(s)}$$

Rearrange the mechanical equation to solve for $\Omega(s)$:

$$\Omega(s) = \frac{k\phi \cdot I(s)}{sJ_m + B_m}$$

Substitute this $\Omega(s)$ back into the electrical equation:

$$U_{an}(s) = I(s)(R_a + sL_a) + k\phi \left(\frac{k\phi \cdot I(s)}{sJ_m + B_m} \right)$$

Factor out $I(s)$ and find a common denominator:

$$U_{an}(s) = I(s) \left[\frac{(R_a + sL_a)(sJ_m + B_m) + (k\phi)^2}{sJ_m + B_m} \right]$$

Divide to get the final transfer function:

$$\frac{I(s)}{U_{an}(s)} = \frac{sJ_m + B_m}{J_m L_a s^2 + (J_m R_a + B_m L_a)s + B_m R_a + (k\phi)^2}$$

Thermal Analysis

Power loss at nominal load:

$$P_{loss} = P_{en} - P_{mn} = (24 \cdot 0.227) - 3.28 = 2.168 \ W$$

The temperature rise (ΔT) between the winding and ambient:

$$\Delta T = P_{loss} \cdot (R_{th:h,a} + R_{th:w,h}) = 2.168 \cdot (29.8 + 5.5) = 76.53 \ K$$

If the max winding temperature is $125 \ ^\circ C$, the ambient temperature should not exceed:

$$T_{ambient_max} = 125 - 76.53 = 48.47 \ ^\circ C$$

MATLAB Representation

```
%%1st assignment modelling
%%NAME: Nmesoma Osakwe, NEPTUN: F8F29H
%%DC Motor Modelling
syms s
%DC MOTOR PARAMTERS
V_an = 24; %V, Nominal Voltage
R_a = 37.5; %Ohm arm. res.
L_a = 0.00137; %H, arm. ind
kfi = 0.0225; %Vs/rad, torque const.
I_anl = 0.0168 %A , no load current
B_m = (kfi^2*I_anl)/(V_an-R_a*I_anl); %Nms, viscous fr. coeff
```

```
J_m = 1.16*10^-7; %kgm2, inertia
%Load parameters
T_nl= 0; %Nm no-load T.
%T_1 = 5*10^-3; %Nm, between n-1 and nominal T
T_n = 4.76*10^-3; %Nm nominal T
%T_2 = 50*10^-3; %Nm, between nominal and stall T
T_s= 14.4*10^-3  %Nm, stall T
%Transfer function for the DC motor
Numerator = [kfi];
Denominator = [L_a*J_m* + (L_a*B_m + J_m*R_a) + R_a*B_m+kfi^2];
%TF HARD CODED
num = [kfi];
denom = [(J_m*L_a) (B_m*L_a+J_m*R_a) (kfi^2+B_m*R_a)];
SYS1 = tf (num, denom)
%TF using feedback comnand
Num1 = [1];
eDen = [L_a R_a];
mDen = [J_m B_m];
TFe = tf (Num1, eDen)
TFm = tf(Num1, mDen)
kfi1 = tf(kfi)
W1 = TFe * kfi1 * TFm;
W2 = kfi1;
SYS2 = feedback (W1, W2)
%TF from Simulink model, using Model Linearizer (ss - state-space)
ss = linsys1;
SYS3 = tf(ss);
SYS3
%residue command - partial fraction expression
[r, p, k] = residue(num, denom)
%residue command - partial 24v step input
num_step = V_an * num;
denom_step = [denom, 0];
[r1, p1, k1] = residue(num_step, denom_step)
```

Transfer Function Results

```
SYS1 =

                 0.0225
  ------------------------------------
  1.589e-10 s^2 + 4.35e-06 s + 0.0005199
```

```
SYS2 =

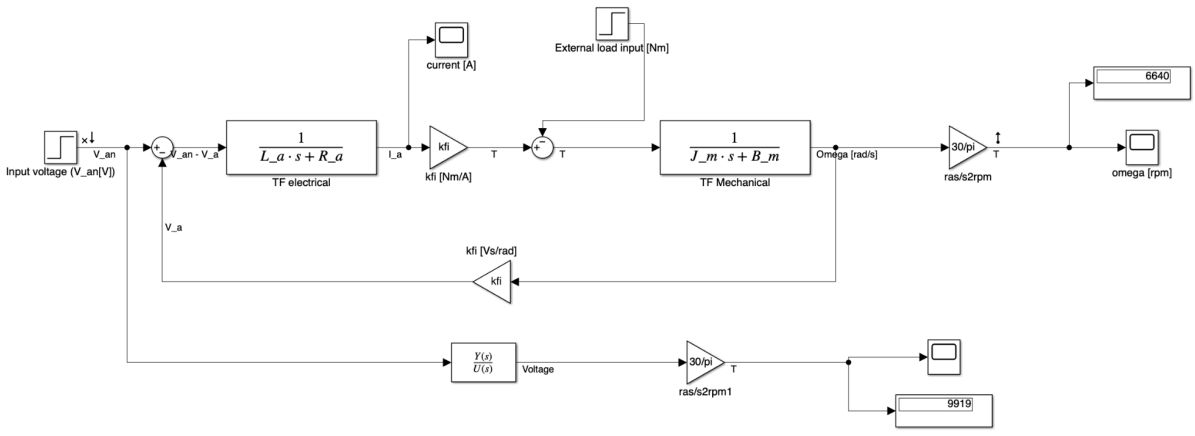
                    0.0225
  ----------------------------------------
  1.589e-10 s^2 + 4.35e-06 s + 0.0005199
```

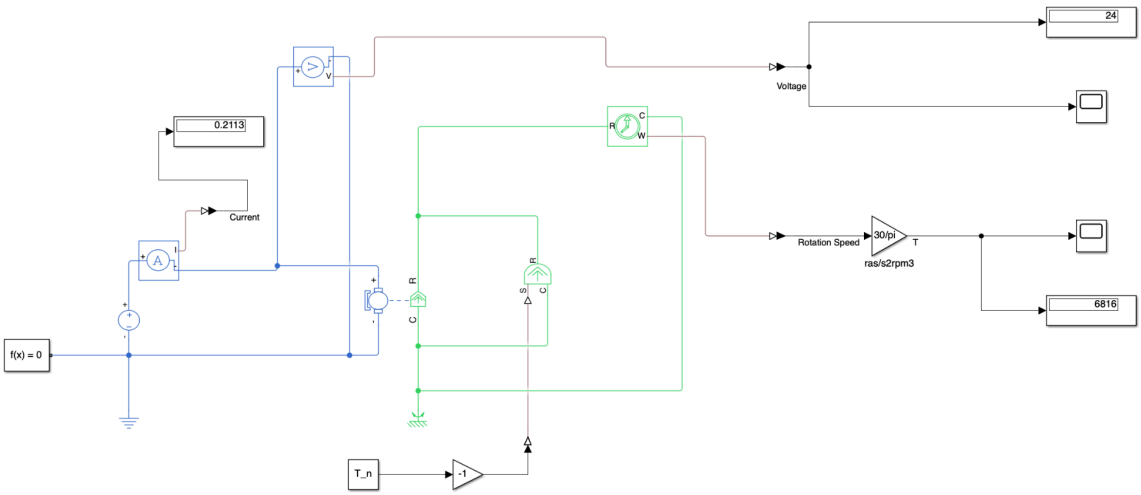
```
SYS3 =

  From input "V_an" to output "ras/s2rpm/T":
           1.352e09
  ---------------------------
  s^2 + 2.738e04 s + 3.271e06

Name: Linearization at model initial condition
Continuous-time transfer function.
```

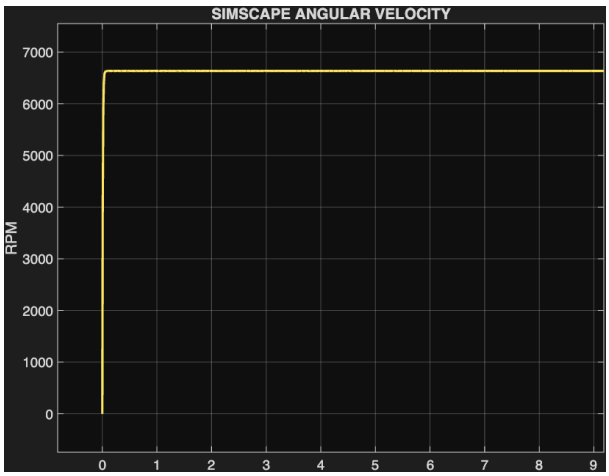
DC Motor Model (SIMULINK/SIMSCAPE)

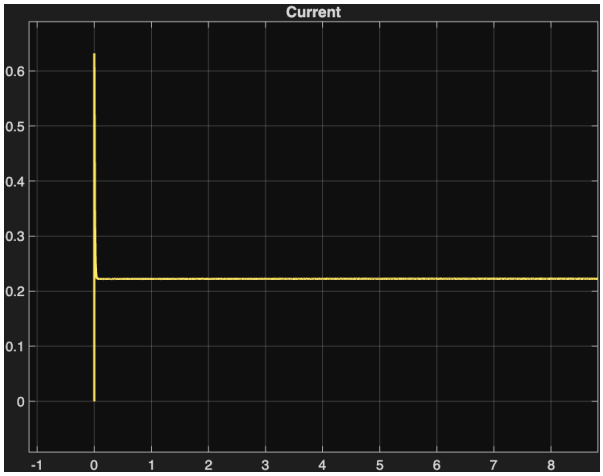




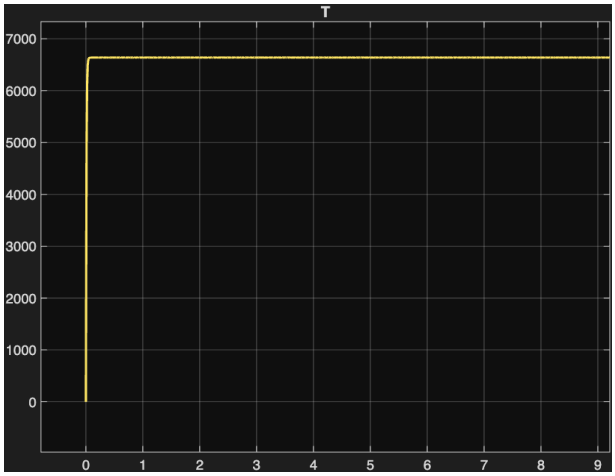
Graphs and Results

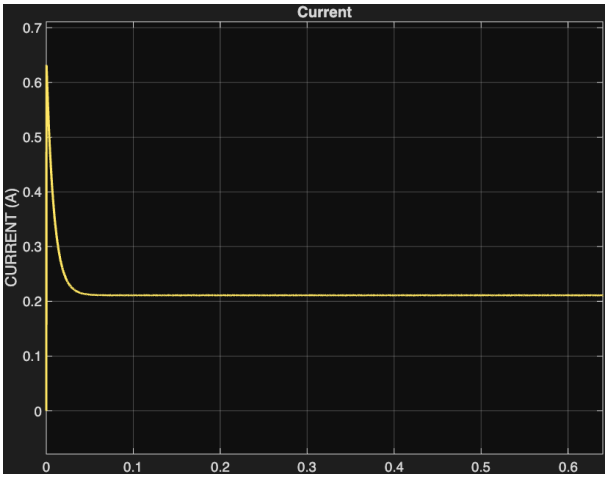
NOMINAL LOAD (SIMSCAPE MODEL) ANGULAR VELOCITY AND CURRENT



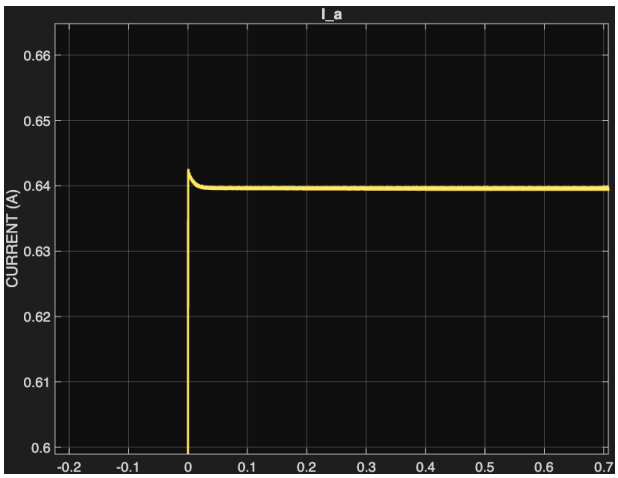


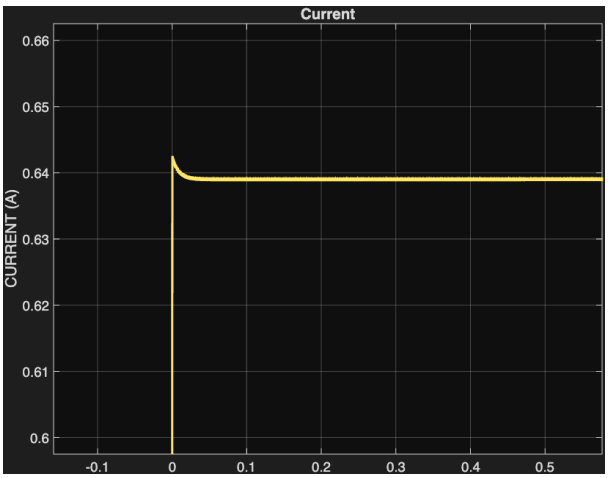
NOMINAL LOAD (BLOCK DIAGRAM) ANGULAR VELOCITY AND CURRENT



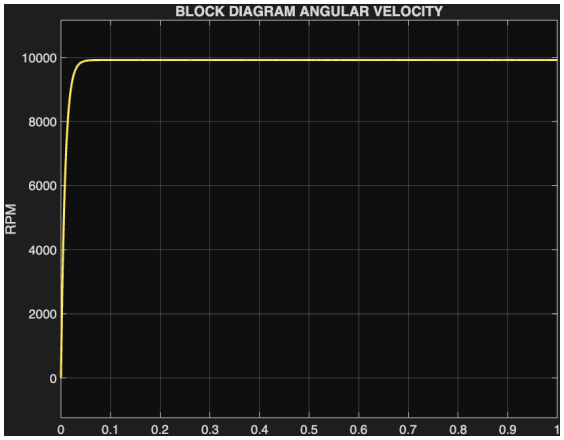


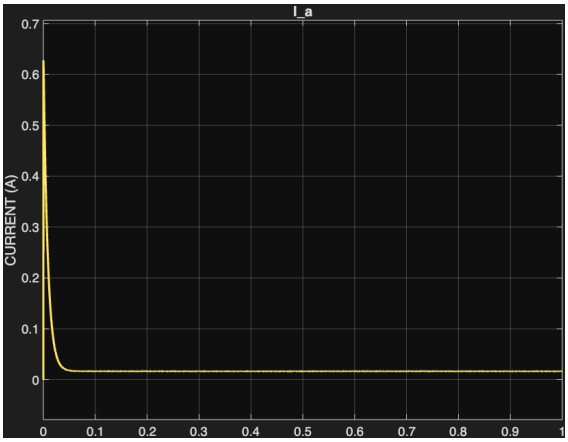
STALL CURRENT(BLOCK DIAGRAM) AND STALL CURRENT(SIMSCAPE MODEL)



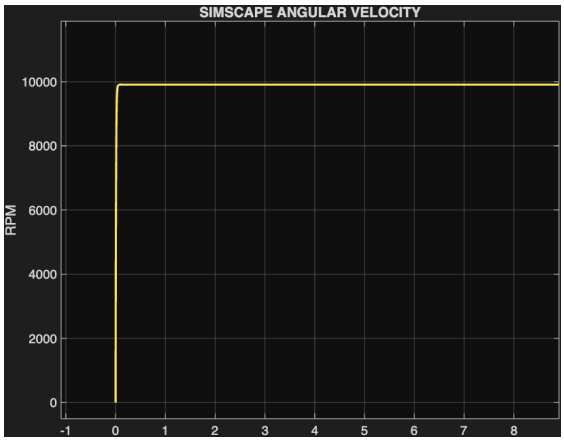


NO LOAD (BLOCK DIAGRAM) ANGULAR VELOCITY AND CURRENT





NO LOAD (SIMSCAPE MODEL) ANGULAR VELOCITY AND CURRENT



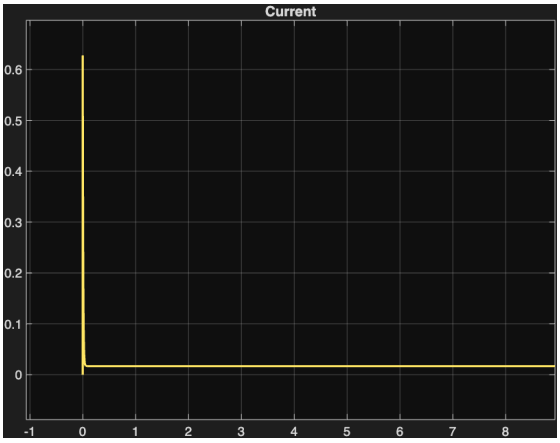


Table Of Comparison

Parameter	Datasheet Value	Simulink	Simscape	Unit
Nominal Voltage	$24V$	$24V$	$24V$	V
No-Load Speed (ω_0)	9920	9919	9909	rad/s (or RPM)
No-Load Current (I_0)	16.8	16.4	16.77	mA
Nominal Speed (ω_{nom})	6580	6640	6637	rad/s (or RPM)
Nominal Current (I_{nom})	0.227	0.2227	0.2226	A
Stall Torque (τ_s)	14.4	-	-	mNm
Stall Current (I_s)	0.64	0.6398	0.6394	A

Summary

In this assignment, a comprehensive mathematical and simulation-based analysis was conducted on the Maxon DC-max 16 S (24V) graphite-brushed DC motor. First, the datasheet parameters were evaluated, allowing for the calculation of missing physical constants such as viscous friction ($B_m \approx 3.639 \times 10^{-7}$ Nms) and the theoretical mechanical/electrical time constants. Next, utilizing Kirchhoff's voltage law and Newton's second law, the exact s-domain transfer functions for angular velocity and armature current were derived and successfully

verified using MATLAB's control system toolboxes.
Finally, the theoretical model was validated by constructing three separate simulations in Simulink (Transfer Function, Block Diagram, and Simscape Physical blocks). The simulation data was logged and compared against the official manufacturer's datasheet. The results showed near-perfect alignment in steady-state metrics, proving that the derived linear approximations and the block models accurately represent the real-world dynamics of the chosen DC motor. Minor discrepancies in the nominal operating points are negligible and can be attributed to the non-linear nature of real-world friction and thermal effects not captured in the ideal linear model. It is also important to note that the electrical time constants were not given in the catalogue; they were calculated based on the other values.